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Polymetallic Nodule Resources: Pacific (Americas and Hawaii)

Executive Summary. Abyssal polymetallic nodules – potato-sized iron–manganese concretions enriched in Ni, Cu, Co and rare-earths – blanket several Pacific basins. The largest field is the Clarion-Clipperton Fracture Zone (CCZ, ~9×10^6 km² between ~5–20°N, 115–160°W)【95†L293-L301】. Here nodules average ~15 kg/m² (locally up to 75 kg/m²)【95†L293-L301】, with typical grades of ~1–1.5% Ni, ~1% Cu, ~0.2–0.3% Co plus ~25–30% Mn【43†L310-L314】【83†L82-L87】. A second major area is the Peru Basin (off Peru, ~14°S, 80–85°W)【33†L198-L202】【95†L303-L305】, roughly half the CCZ size, with ~10 kg/m² nodules【95†L303-L305】. Notable planned fields include a US-study target south of Hawaii (US EEZ)【83†L82-L87】. Geologically, nodules occur on flat abyssal plains (4–6 km depth【33†L211-L215】【81†L90-L94】) with very low sedimentation, high oxygen bottom-water and moderate productivity (e.g. Equatorial Pacific upwelling fuels metal remobilization)【97†L19-L27】【81†L169-L174】. Environmental protections are extensive: for example ISA-designated CCZ "Areas of Particular Environmental Interest" (APEIs, ~2.0×10^6 km² total) prohibit mining【51†L234-L239】, and national MPAs (US Papahānaumokuākea ~1.5×10^6 km²【53†L181-L188】, Pacific Remote Islands ~1.27×10^6 km²【100†L169-L173】; Chile's Nazca-Desventuradas ~3.0×10^5 km²【58†L87-L90】; Ross Sea ~1.55×10^6 km²【64†L87-L90】) also cover key abyssal habitats. Under UNCLOS the deep-sea "Area" (beyond national EEZs) is managed by the International Seabed Authority (ISA)【88†L193-L201】. To date ISA has issued numerous 15-year exploration contracts (CCZ focus: India, Japan, China, Russia, France, Germany, UK, Belgium, Nauru, Tonga, Kiribati, Singapore, Cook Islands, Jamaica, etc【71†L232-L239】【72†L327-L331】). No deep nodules have yet been mined, and key data gaps remain: most geologic and ecological information is from CCZ, while many basins (e.g. US Pacific, southern Pacific) lack baseline surveys【83†L93-L100】【95†L331-L338】.

Known Occurrences and High-Potential Areas

Major nodule fields lie on abyssal plains from the Arctic to Antarctic Pacific. **Clarion-Clipperton Fracture Zone (CCZ)**: spanning roughly 9–20°N, 115–160°W (between Mexico and Hawaii), the CCZ is the world's largest nodule province【95†L293-L301】. Average nodule abundance is ~15 kg/m² (locally up to ~75 kg/m²)【95†L293-L301】, yielding massive metal resources. Nodules here (and generally) are ~25–30% Mn, 6–8% Fe, with ~1.2–1.5% Ni, ~1.0–1.4% Cu, 0.2–0.3% Co【43†L310-L314】. Many ISA exploration contracts overlap the CCZ (sponsored by India, Japan, China, Russia, France, Germany, UK, Belgium, Nauru, Tonga, Kiribati, Singapore, Cook Islands, Jamaica, etc.【71†L232-L239】【72†L327-L331】). **Peru Basin (equatorial SE Pacific)**: centered ~10–17°S, 80–90°W off Peru, this basin (~5×10^6 km²) contains abundant nodules (~10 kg/m²)【95†L303-L305】. High equatorial productivity and diagenetic input (near oxygen-minimum waters) favor nodule growth here【8†】. No ISA lease is known here; any future exploration would fall under Peru/Chile EEZ law. **Hawaiian EEZ (Central Pacific)**: US waters south of Hawai'i (~20°N, 155°W) include abyssal plains (4–6 km depth) targeted by planned BOEM/USGS research【83†L82-L87】. These nodules are expected to have grades similar to CCZ (Ni, Cu, Co, REE【83†L82-L87】). Other Pacific areas (e.g. near US Pacific Remote Islands, or off Chile/Antarctica) may host nodules but are largely uncharacterized.

【28†embed_image】 *Figure 1: Distribution of major polymetallic nodule fields in the Pacific. The Clarion-Clipperton Zone (CCZ, shaded) extends from off Mexico (~10°N) to near Hawai'i, with additional fields in the Peru Basin (~15°S)【95†L293-L301】【95†L303-L305】. (Source: adapted from USGS【26†L13-L15】).*

Geological & Sedimentary Setting

Nodules form on flat, stable abyssal plains (typically 3.5–6 km depth【33†L211-L215】【81†L90-L94】) under very low sedimentation (mm per millennium)【33†L211-L215】. Sediments are fine-grained pelagic clays and oozes (e.g. red clay, diatomaceous ooze) often saturated with pore water. Key conditions include **very low sedimentation** (preventing nodule burial) and strong abyssal bottom currents (especially Antarctic Bottom Water) that winnow fines【97†L19-L27】. In practice, abyssal Pacific waters flush away clays and keep nodules at the sediment surface【97†L19-L27】. The **bathymetry** of the CCZ and Peru Basin is typical abyssal plain (~4–5 km depth). For example, CCZ samples are found at ~4,000–5,500 m【33†L211-L215】; Smithsonian notes nodule depths ~3,960–5,790 m【81†L90-L94】. These plains are underlain by old oceanic crust; nodules preferentially accumulate where nearby topography is smooth (no seamounts or trenches to disturb deposition).

Oceanographic Controls

Ocean chemistry and circulation influence nodule formation. Surface productivity supplies metals: sinking biogenic detritus (plankton carcasses) delivers Fe/Mn to the seafloor【81†L169-L174】. Dissolved Mn/Fe in bottom waters precipitate onto nodule "seeds" via hydrogenous processes, while diagenesis in sediments also releases metals into pore water. Pacific deep water (ventilated by Antarctic and Pacific sources) remains well-oxygenated, supporting slow hydrogenetic growth. An important control is the **Antarctic Bottom Water** flow across the Pacific abyssal plains. This "geyser" of cold, oxygen-rich water flushes fine sediment away and delivers dissolved oxygen, preventing anoxia and rapid burial【97†L19-L27】. Oxygen minimum zones (OMZs) impinge on the upper slopes (e.g. off Peru), aiding diagenetic nodule growth there. Overall, Pacific nodules thrive where organic deposition is moderate and sediment supply is minimal (south of Hawai'i, equatorial high-productivity belts, etc).

Metal Grades

Polymetallic nodules are enriched in battery metals. Typical nodule composition (CCZ-level) is ~27–30% Mn (as mixed oxides) and ~6–8% Fe【43†L310-L314】. Valuable metals occur at ~1–2%: Ni ~1.2–1.5 wt%, Cu ~1.0–1.4 wt%, Co ~0.2–0.25 wt%【43†L310-L314】. For economic planning, ISA notes that nodules with mass >5 kg/m^2 and **combined** Ni+Cu+Co >2% are considered viable【33†L221-L224】. Rare earth elements (REE+Y) occur at ~0.2–0.5% in CCZ nodules (including Y up to ~0.1%【83†L84-L87】). Metal grades vary by field (e.g. Peru Basin nodules are often higher in Mn/Fe due to diagenesis, whereas others have relatively more hydrogenous Ni/Cu). **Table 1** compares key fields:

Field	Location (lat, lon)	Depth (m)	Ni wt%	Cu wt%	Co wt%	Exploration / Status
Clarion-Clipperton	≈10–20°N, 115–160°W (Pacific off Mexico) 【95†L293-L301】	4000–5500	~1.2	~1.1	~0.25	Multiple ISA contracts (e.g. India, Japan, China, Russia, France, Germany, UK, Belgium, Nauru, Tonga, Kiribati, Singapore, Cook Isl., Jamaica…) 【71†L232-L239】 【72†L327-L331】; extensive research.
Peru Basin	≈10–17°S, 80–90°W (off Peru) 【33†L198-L202】 【95†L303-L305】	~4000–5000	~1.0*	~0.9*	~0.2*	Recognized abundant nodules 【33†L198-L202】 (~10 kg/m^2 【95†L303-L305】). No ISA lease; any mining would need Peruvian/Chilean EEZ approval.
Hawaiian (US EEZ)	≈18–23°N, 155–160°W (south of Hawai'i) 【83†L82-L87】	4000–6000	~1.1	~1.0	~0.2	Not contracted (US not in ISA). BOEM/USGS planning resource surveys 【83†L82-L87】. Possible future US domestic leasing (Deep Seabed Hard Mineral Resources Act).
Typical nodule	–	–	1.2–1.5	1.0–1.4	0.2–0.25	–

Field	Location (lat, lon)	Depth (m)	Ni wt%	Cu wt%	Co wt%	Exploration / Status

Table 1. Comparison of major Pacific nodule fields (depth and metal grades are typical for each region). **Sources:** ISA/USGS data【71†L232-L239】【72†L327-L331】; WOR (CCZ area, densities)【95†L293-L301】【95†L303-L305】. (*Estimated; see text.*)

Exploration and Licensing

Under UNCLOS, nodule mining in international waters is managed by the **International Seabed Authority** (ISA)【88†L193-L201】. To date ~15-year exploration contracts have been granted for nodules in the CCZ (and one "Western Pacific" block). Initial leases (2001–2004) went to India, Japan, France, Russia, China, Interoceanmetal (Bulgaria/Cuba/Poland etc) and S. Korea【71†L232-L239】. After 2005, 11 more nodule contracts were issued (e.g. Germany/BGR, Nauru Ocean Resources Inc., Tonga Offshore Mining, Global Sea Minerals (Belgium), UK Seabed Resources I/II, Marawa (Kiribati), Ocean Mineral Singapore, Cook Islands, China Minmetals, Pioneer (China), Jamaica)【71†L232-L239】【72†L327-L331】. Each contract covers up to 75,000 km^2. (Fig. 2 below charts the timeline of these awards.) In national waters (EEZs), coastal states control mineral rights: e.g. the US currently limits nodule research (via BOEM studies【83†L82-L87】) and has no active leasing program; Peru/Chile would have jurisdiction over their margin nodules (no known projects announced); similarly Mexico, Canada, etc.

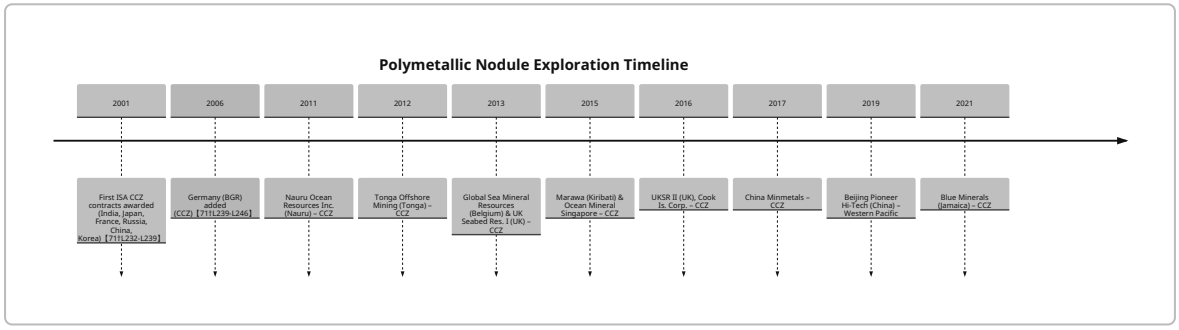


Figure 2: Timeline of ISA polymetallic nodule exploration contracts (sponsor country and license area). Early contracts (2001–2004) established the CCZ mining regime【71†L232-L239】; newer entries include Kiribati, Singapore, China Pioneer (Western Pacific) and Jamaica【72†L327-L331】.

Environmental Constraints and Protected Areas

Deep-seabed mining raises major environmental concerns. In the CCZ, the ISA's Environmental Management Plan designates **13 no-mining Areas of Particular Environmental Interest** (APEIs; ~1.97×10^6 km² total)【51†L234-L239】 to conserve biodiversity. Nationally-designated MPAs also cover extensive deep-sea habitats: for example, the U.S. Papahānaumokuākea Marine National Monument (northwestern Hawaiian EEZ) spans ~1.5×10^6 km²【53†L181-L188】, and the Pacific Remote Islands Monument (south/west of Hawai'i) ~1.27×10^6 km²【100†L169-L173】, both with strict bans on extractive activities. Chile created the Nazca-Desventuradas Marine Park (~298,000 km², fully no-take) around San Félix/San Ambrosio islands【58†L87-L90】. The Ross Sea Marine Protected Area (~1.55×10^6 km², ~1.12×10^6 km² fully no-take) protects Southern Ocean ecosystems【64†L87-L90】. These and other ocean reserves (e.g. Galápagos, Antarctic MPAs) mean that large portions of the Pacific are off-limits to mining. In practice, any nodule mining proposal must navigate both international regulations and coastal-state laws (EEZ minerals belong to the coastal nation).

Data Gaps and Uncertainties

Despite decades of study, many uncertainties remain. Vast areas of potential nodule terrain have never been surveyed in detail: most data come from a few CCZ sites (and some small surveys off Peru), whereas regions like the U.S. Pacific EEZ or sub-Antarctic Pacific lack baseline information【83†L93-L100】. Ecological understanding is especially thin: benthic communities on nodules may take centuries to recover【83†L93-L100】, yet species composition is known for only a fraction of fields (the CCZ has ~600 described nodule-associated species【81†L194-L202】). Resource estimates likewise have wide error bars; even CCZ tonnage (e.g. "21 billion tonnes nodule mass"【95†L293-L301】) is subject to uncertainty. Crucially, no commercial-scale mining has occurred, so the technical and environmental feasibility remains untested. As one review notes: *"Whether, and how much metal can be obtained from manganese nodules in the future is completely uncertain."*【95†L331-L338】. Closing these gaps will require coordinated mapping, sampling and impact studies across the Pacific basins.

Sources: This report synthesizes peer-reviewed studies, ISA documents and government reports on Pacific polymetallic nodules【33†L198-L202】【51†L234-L239】【95†L293-L301】【83†L82-L87】. Tables and figures integrate data from these sources as cited. All exploration and reserve information is from authoritative ISA and national agency publications【71†L232-L239】【64†L87-L90】. Speculative statements are flagged or avoided.
